

Junior Cloud Security Engineer

Internship Task Report

1.0 Task Overview

Task ID	CSE-AWS-T-001-02
Task Title	Introduction to Cloud Security Principles and Shared Responsibility Model
Skill Domain	AWS Cloud Security Fundamentals
Difficulty	Beginner
Duration	10 Hours
CPE Credits	2 CPE Credits
Cloud Platform(s)	AWS Free Tier (IAM, S3, EC2, CloudTrail)
AWS Region(s) Used	Us-east-2c
Primary Tools	AWS Management Console, AWS CLI, ChatGPT or Claude AI, Draw.io, Excel/Google Sheets
Target Environment	AWS Free Tier (IAM, S3, EC2, CloudTrail),TryHackMe, Powned
MITRE ATT&CK IDs	MITRE ATT&CK T1087 (Account Discovery), T1530 (Data from Cloud Storage Object)
Guiding Frameworks	ISO 27001: Controls A.5.1 (Information Security Policies), A.9 (Access Control), A.12 (Operations Security), SFIA, PR-CDA-001 (Cloud Security Analyst), PR-INF-001 (Infrastructure Security Specialist)
Regulatory Alignment	ISO 27001: Controls A.5.1 (Information Security Policies), A.9 (Access Control), A.12 (Operations Security), SFIA, PR-CDA-001 (Cloud Security Analyst), PR-INF-001 (Infrastructure Security Specialist)

1.1 Introduction

The project is focused on AWS Shared Responsibility Model. With hands-on AWS configuration, threat modeling and cloud security labs, it highlights security risks such as misconfigurations of S3 buckets, overprivileged accounts, or not enabling MFA. Gaining knowledge on these skills will help cloud engineers to identify misconfigurations,that further reduce likelihood of security risk.

1.2 Objectives & Learning Outcomes

- **Objective 1:** Understand the core principles of cloud security and how they relate to the AWS Shared Responsibility Model.
- **How I met it:** I studied AWS Shared Responsibility Model, also differentiate between IaaS, PaaS, SaaS services. Created Diagram that describes responsibilities of Cloud Provider and customers.
- **Objective 2:** Configure your AWS Free Tier environment for observation.
- **How I met it:** I configured an AWS Free Tier environment and created an IAM user with least privilege permissions, enabling AWS CloudTrail logging, listing S3 buckets using the AWS CLI, and launching an Amazon EC2 Free Tier instance.
- **Objective 3:** Map each service's security responsibilities between AWS and the customer.
- **How I met it:** I created a responsibility matrix for IAM, Amazon S3, Amazon EC2, and AWS CloudTrail, defining AWS responsibilities, customer responsibilities, and potential security risks for each service.
- **Objective 4 :** Develop a basic threat model illustrating how misconfigurations could lead to compromise.
- **How I met it:** I developed a risk matrix and a threat model diagram to identify common cloud misconfigurations, attack paths, their potential impacts, and recommended mitigation strategies.
- **Objective 5:** Reinforce learning through guided labs simulating real-world scenarios.
- **How I met it:** completed the TryHackMe Security Principles lab and the PwnedLabs Breach in the Cloud lab.

1.3 Scope & Legal Authorization

Authorized environments used:

- Personal AWS Free Tier account — Account ID ending in 4456 — us-east-2c region
- TryHackMe: Security Principles Lab — Completed within the authorised TryHackMe training environment
- PwnedLabs: Breach in the Cloud — Assume Privileged Role with External ID — Personal VM
- Draw.io — Used to create the AWS Shared Responsibility Diagram and the Threat Model Diagram
- Microsoft Excel — Used to create the AWS Responsibility Matrix and Risk Matrix
- Microsoft Word — Used to prepare the project documentation and final report
- AI ChatGPT : assistance in some phases (for example, refining descriptions)

All practical activities were performed only within authorized training environments, including my personal AWS Free Tier account, TryHackMe, and PwnedLabs. No production AWS accounts, third-party cloud environments, or real customer data were accessed during this project. AI assistance (ChatGPT) was used only to improve explanations, refine documentation, and enhance report clarity. All AWS configurations, laboratory activities, screenshots, and technical evidence were completed and verified manually by the author.

1.4 MITRE ATT&CK Cloud Threat Mapping

MITRE ATT&CK Technique ID	Technique Name	Relevance to This Task
T1078	Valid Accounts	Creating IAM users and understanding how attackers can misuse credentials to gain unauthorized access.

MITRE ATT&CK Technique ID	Technique Name	Relevance to This Task
T1087.004	Cloud Account Discovery	IAM users, roles, and permissions were reviewed to understand account management and how attackers may enumerate cloud identities after gaining access.
T1530	Data from Cloud Storage Object	Examination of Amazon S3 bucket, what would be risk if publicly accessible.
T1098	Account Manipulation	Least-privilege IAM configuration and permission management demonstrated how excessive permissions or unauthorized account changes can increase security risk.
T1562.001	Impair Defenses: Disable or Modify Security Tools	AWS CloudTrail logging was enabled to prevent attackers from hiding their activities by disabling or modifying audit logs.
T1190	Exploit Public-Facing Application	The threat model included EC2 security group misconfigurations that could expose public-facing services to exploitation.

2.0 Environment & Tool Setup

2.1 Cloud Architecture Overview

[Describe your cloud lab environment. Complete the table below for every resource deployed. Add or remove rows to match your specific task. Include both the AWS services you configured and any local tools used alongside them.]

Cloud Resource / Component	Type / Service	Role in Task	Region / Config Notes
AWS IAM User (test-user)	AWS Identity and Access Management (IAM)	Created a test user with least privilege permissions to demonstrate secure identity	AWS Free Tier account / MFA not enabled (lab environment) / Read-only permissions assigned.
CloudTrail	AWS CloudTrail	Enabled logging to record AWS API activity for auditing, monitoring, and incident investigation.	Multi-region enabled / S3 log bucket
Amazon EC2 Instance	Amazon EC2	Free Tier instance to understand compute security, security groups, and cloud infrastructure	Enabled in us-east-2c/ t2.micro / Amazon Linux.

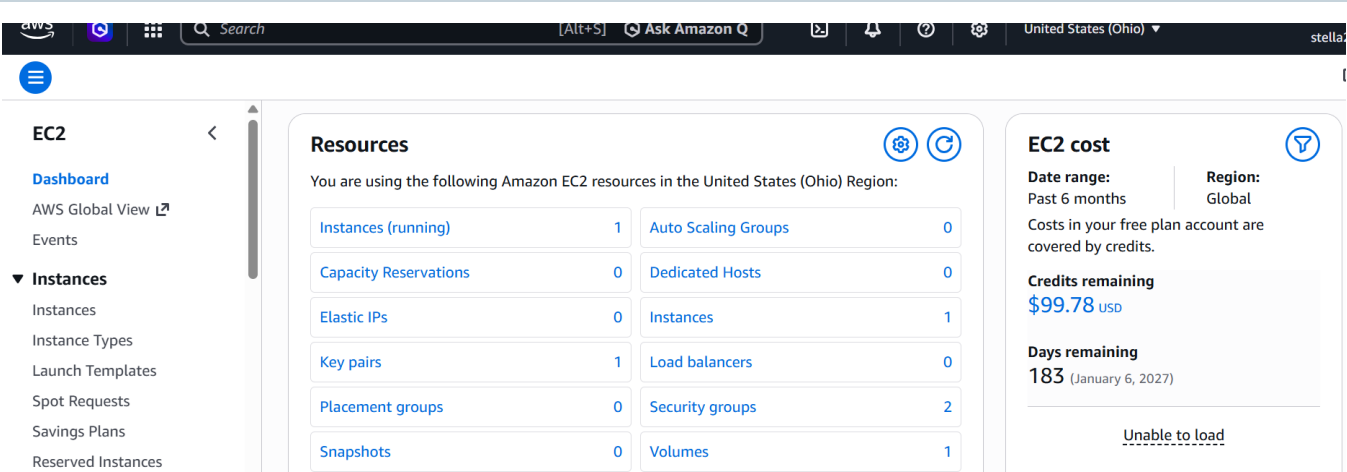
Cloud Resource / Component	Type / Service	Role in Task	Region / Config Notes
Amazon S3 Bucket	Amazon S3	Used for CloudTrail log storage and to study cloud storage security and shared responsibility.	Private / versioning enabled
AWS CLI	Local Command-Line Tool	Used to configure AWS credentials and interact with AWS services, including listing S3 buckets.	Installed on Window Machine
Kali Linux Virtual Machine	Operating System	Used as the primary environment for AWS CLI commands and completing the PwnedLabs investigation.	VMware Workstation virtual machine.
Draw.io	Diagramming Tool	Used to create the Shared Responsibility Diagram and the Threat Model Diagram	Web Browser
Microsoft Excel	Spreadsheet Software	Used to create the Responsibility Matrix and Risk Matrix.	Local application.
Microsoft Word	Word Processing Software	Used to prepare the project documentation and final report.	Local application.

AWS free Tier environment used for Amazon EC2 instance:

The screenshot shows the AWS Management Console interface for the EC2 service. The main content area displays a table of instances with the following details:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
PhaseB-EC2	i-08964f8ba50b4c8a9	Stopped	t3.micro	-	us-east-2c

The left sidebar contains the navigation menu, and the top bar shows the user's profile and the current region (United States (Ohio)).



2.2 AWS CLI & Tool Configuration

[For each primary tool used, document how you installed and configured it. Use the table below — copy and add rows as needed. Focus on configuration choices that were specific to this task (e.g. which AWS profile, region, or access key type used).]

Tool / Service	Version	Installation Method	Purpose in This Task
AWS CLI	version 2x	Installed on Kali Linux using the official AWS CLI installation package. Configured using aws configure with AWS Access Key ID, Secret Access Key, region, and output format. Installed on window by downloading exe file	Used to configure AWS credentials, list Amazon S3 buckets, and interact with AWS services from the command line
AWS Management Console	Web Console	No installation required; accessed through a web browser using an AWS Free Tier account	Used to create IAM users, enable AWS CloudTrail, launch an Amazon EC2 instance
Kali Linux	2025.x	Installed as a VMware virtual machine	Used as the primary operating system for running AWS CLI commands and completing Pwned cloud security labs
jq	Version 1.6	Installed using sudo apt install jq -y	Used to parse and format JSON output during the PwnedLabs "Breach in the Cloud" investigation
Draw.io	Web Browser	Accessed through the Draw.io web application	Used to create the AWS Shared Responsibility Diagram and the Threat Model Diagram.
Microsoft Excel	Microsoft 365	Installed locally on Windows	Used to create the AWS Responsibility Matrix and Risk Matrix

Tool / Service	Version	Installation Method	Purpose in This Task
Microsoft Word	Microsoft 365	Installed locally on Windows	Used to prepare the project report and supporting documentation

Key Configuration Commands Used

AWS CLI Configuration: aws configure

Output:

AWS Access Key ID: *****

AWS Secret Access Key: *****

Default region name: us-east-2c

Default output format: json

Verify AWS Identity : aws sts get-caller-identity

Output:

```
{
  "UserId": "AIDAXXXXXXXXXXXXXX",
  "Account": "[REDACTED]",
  "Arn": "[REDACTED]"
}
```

Verify AWS CLI Installation: aws --version on windows command prompt

Output:

aws-cli/2.35.16 Python/3.14.5 Windows/11 exe/AMD64

for kali :

aws-cli/2.xx.x Python/3.xx.x Linux/x86_64

Verify jq Installation: jq --version

Output:

jq-1.6

Verify Connectivity: aws iam list-users

Output:

```
{
```

```
"Users": [  
  {  
    "UserName": "test-user"  
  }  
]  
}
```

2.3 AI Tool Usage Declaration

[Declare any AI tools used during this task. Be specific about what they were used for. If no AI tools were used, write: No AI tools were used in this task.]

AI Tool	Used For	What Was NOT Done with AI
ChatGPT / Claude	Explaining IAM policy syntax; summarizing CloudTrail event structures; reviewing risk matrix language. Explaining AWS Shared Responsibility concepts. Refining report wording and grammar. Improving descriptions in the responsibility matrix and risk matrix. Generating an analogy comparing on-premises and cloud responsibilities; final report.	AI did not generate technical findings, execute CLI commands, produce screenshots, or create any evidence files. All AWS configurations, commands, screenshots, and findings were completed and verified by the author.

All technical findings, CLI outputs, configurations, and evidence in this report are the intern's own work.

3.0 Methodology — Phase Execution

Phase A: Theory & Conceptual Foundation

Phase Objective	<i>Understand the core principles of cloud security and how they relate to shared responsibility.</i>
Estimated Time	~90 minutes
Evidence File(s)	diagram (PhaseAshared_responsibility_diagram.png) and summary (phaseA_summary.pdf).

Actions Taken

AI Assistance Used

Prompt used:

"Explain the AWS Shared Responsibility Model and how responsibilities differ between IaaS, PaaS, and SaaS. Also provide a simple analogy comparing on-premises infrastructure with cloud responsibilities."

Summary of the AI response (in my own words):

ChatGPT explained that AWS is responsible for securing the cloud infrastructure, while customers are responsible for protecting their data, identities, operating systems, and application configurations. It also explained that customer responsibilities decrease as services move from IaaS to PaaS and SaaS. The apartment rental analogy helped understanding as layman person that AWS maintains the building, while the customer is responsible for securing and managing the contents inside their rented space. I reviewed this explanation alongside the AWS documentation.

Example — CLI command syntax:

```
[aws service command --flag value] # Paste relevant output here — redact sensitive values such as account IDs where necessary
```

[Continue documenting each step in sequence. For console-based actions, describe the navigation path and outcome. For policy edits, paste the JSON block below.]

Key Observations

During this phase, I learned how the AWS Shared Responsibility Model divides security responsibilities between AWS and the customer. AWS is responsible for securing the cloud infrastructure, while customers are responsible for protecting their data, user accounts, applications, and configurations.

I also understood that customer responsibilities change depending on the cloud service model. In IaaS, customers manage more security controls than in PaaS or SaaS. This phase also showed me that many cloud security incidents happen because of customer misconfigurations, such as overly permissive IAM permissions or publicly accessible S3 buckets.

Evidence Produced

- phaseA_summary.pdf : Describes Shared Responsibility Model along with describing how IaaS, PaaS and SaaS are different from each other.
- PhaseAshared_responsibility_diagram.png : Diagram illustrating AWS and customer security responsibilities.

Phase B: Cloud Environment Setup & Target Preparation

Phase Objective	Configure your AWS Free Tier environment for observation.
Estimated Time	~120 minutes
Evidence File(s)	<i>PhaseB_CloudTrail_Dashboard.png, PhaseB_EC2_Instance.PNG, PhaseB_IAM_UserList.png, PhaseB_S3_bucket_CLI.PNG</i>

Actions Taken

I logged in to my AWS Free Tier account using the AWS Management Console. I created a test IAM user with least-privilege permissions by attaching the ReadOnlyAccess policy. This user was created to demonstrate secure access management without granting unnecessary permission.

Next, I enabled AWS CloudTrail to record API activity across all AWS regions. I configured a new S3 bucket to store CloudTrail log files.

I launched one t3.micro EC2 instance under the AWS Free Tier. I also configured the AWS CLI on my Kali Linux virtual machine and window's using my AWS access keys and verified the configuration by checking my AWS identity. Finally, I listed the available S3 buckets using the AWS CLI.

AWS CLI Commands Used:

```
aws configure
```

```
aws sts get-caller-identity
```

```
aws s3 ls
```

Console Navigation:

IAM → Users → Create User → Attach ReadOnlyAccess policy

CloudTrail → Create Trail → Enable Multi-region Trail

EC2 → Instances → Launch Instance → Amazon Linux → t3.micro

S3 → CloudTrail log bucket created automatically during trail setup

Figure 1. IAM test user created with the ReadOnlyAccess policy.

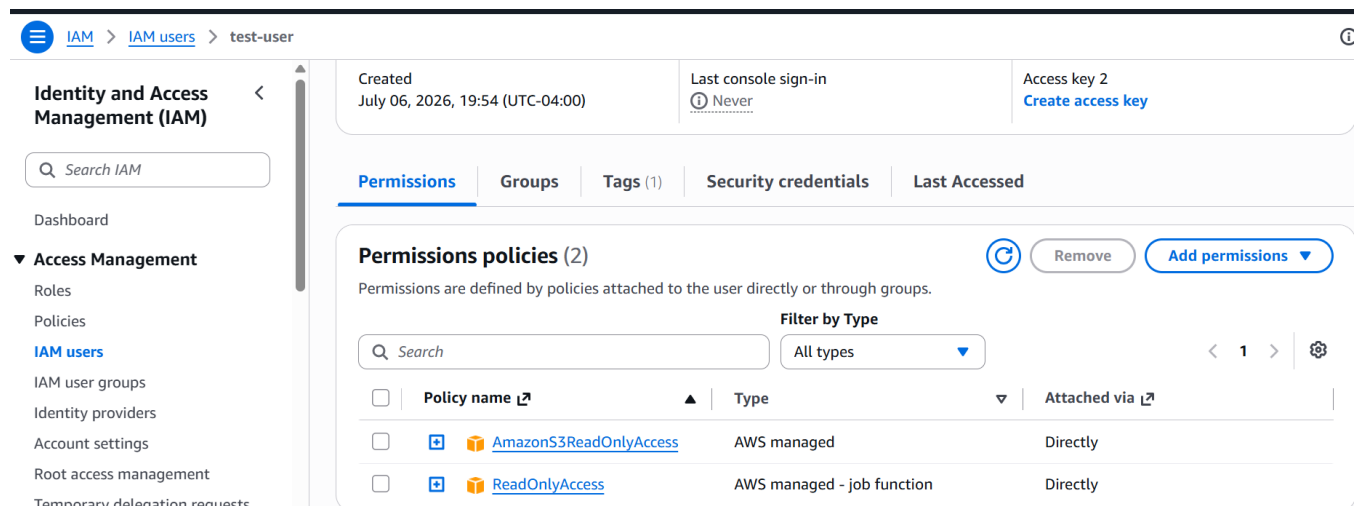


Figure 2. AWS CloudTrail configured for multi-region logging.

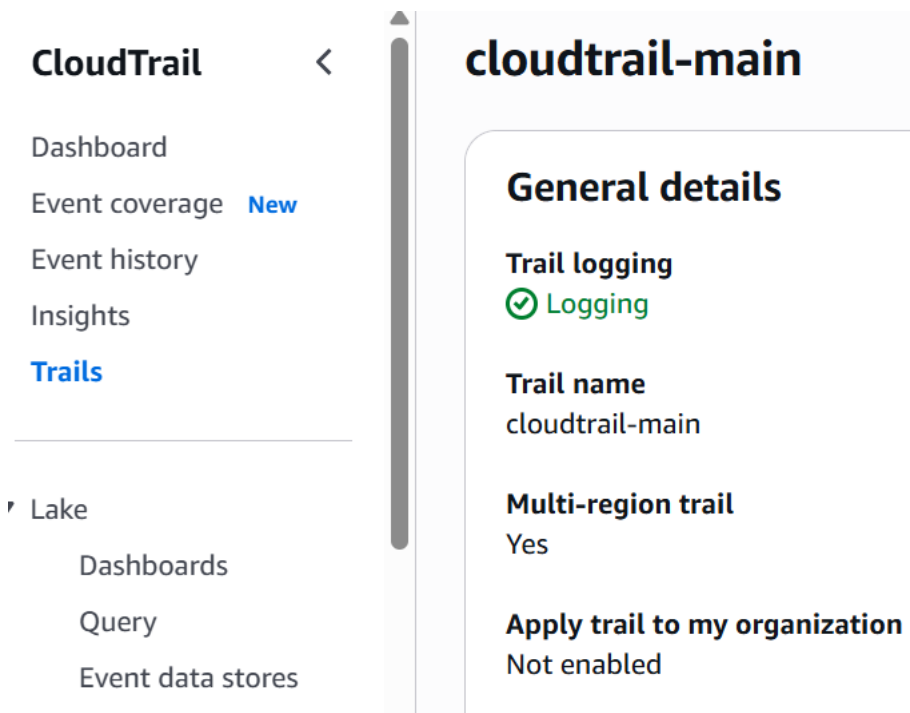


Figure 3. Amazon EC2 t3.micro instance.

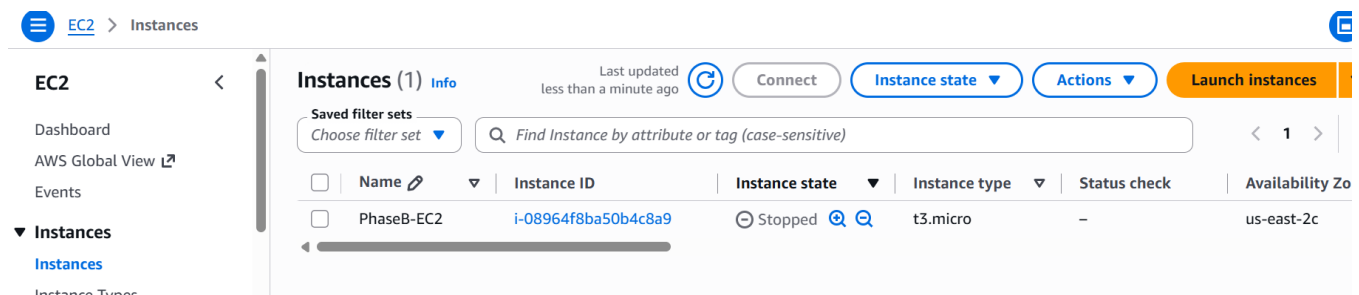


Figure 4. AWS CLI listing available S3 buckets.



Key Observations

The AWS environment was successfully configured, and all required resources were created without major issues. I encountered a few permission-related errors while using the AWS CLI because the test IAM user had least-privilege permissions. For Listing S3 bucket, I had to assign S3ReadOnlyAccess, I was able to complete the required tasks.

Enabling CloudTrail, creating IAM users with limited permissions, and using least privilege helped establish a secure baseline for the environment.

Evidence Produced

- *PhaseB_CloudTrail_Dashboard.png*: Screenshot confirming CloudTrail is enabled.
- *PhaseB_EC2_Instance.PNG*: Screenshot of the EC2 t3. micro instance.
- *PhaseB_IAM_UserList.png*: Screenshot showing the test IAM user with least-privilege permissions.
- *PhaseB_S3_Bucket_CLI.PNG*: WS CLI output showing the available S3 buckets.

Phase C: Core Execution

Phase Objective	Map each service's security responsibilities between AWS and the customer.
Estimated Time	~150 minutes
Evidence File(s)	PhaseC_Responsibility_Matrix.xlsx

Actions Taken

I created a responsibility matrix for four AWS services: IAM, Amazon S3, Amazon EC2, and AWS CloudTrail. For each service, I identified the security responsibilities managed by AWS and those managed by the customer based on the AWS Shared Responsibility Model.

I then documented one common security risk for each service. These included overly permissive IAM permissions, publicly accessible S3 buckets, unpatched EC2 instances, and disabled or unmonitored CloudTrail logging. After completing the matrix, I used ChatGPT to improve the wording of the descriptions while keeping the technical meaning unchanged. I reviewed the final content before adding it to the spreadsheet.

AI Assistance Used

Prompt used

"Refine the AWS responsibility matrix descriptions for IAM, S3, EC2, and CloudTrail to improve clarity while keeping the technical meaning the same."

Summary

ChatGPT improved the wording and readability of the responsibility descriptions. I reviewed the suggestions and ensured they accurately reflected AWS security best practices before including them in the matrix.

Example — CLI command syntax:

```
[aws service command --flag value] # Paste relevant output here — redact sensitive values  
such as account IDs where necessary
```

[Continue documenting each step in sequence. For console-based actions, describe the navigation path and outcome. For policy edits, paste the JSON block below.] N/A

Key Observations

Creating the responsibility matrix helped me understand that cloud security is a shared effort between AWS and the customer. AWS secures the underlying cloud infrastructure, while customers are responsible for configuring services securely and protecting their own data and identities. I also observed that many security incidents can

be prevented by following the principle of least privilege, securing storage resources, keeping systems updated, and monitoring account activity.

Evidence Produced

- PhaseC_Responsibility_Matrix.xlsx: Responsibility matrix for IAM, Amazon S3, Amazon EC2, and AWS CloudTrail.

Figure 5: Responsibility Matrix screenshot

A	B	C	D
Service	AWS Responsibility	Customer Responsibility	Potential Risk
IAM	Maintain the IAM infrastructure, authentication services, and availability.	Create and manage IAM users, groups, and roles based on organizational requirements. Apply the principle of least privilege by granting only the permissions necessary for users to perform their tasks. Enable Multi-Factor Authentication (MFA) for privileged and sensitive accounts, regularly rotate access keys and credentials, monitor IAM activity, and periodically review and remove unnecessary permissions to maintain a secure identity and access.	Granting excessive permissions, failing to enable MFA, using long-lived access keys, or neglecting periodic permission reviews can result in unauthorized access, privilege escalation, and compromise of AWS resources.
Amazon S3	Protect the underlying storage infrastructure, hardware, and service availability.	Customers are responsible for securing S3 buckets by configuring access permissions, enabling encryption where appropriate, and preventing unintended public access.	Publicly accessible S3 buckets, excessive access permissions, or missing encryption can expose sensitive information to unauthorized users, leading to data breaches, data loss, and potential non-compliance with organizational security policies and regulatory requirements.
Amazon EC2	Secure the physical servers, virtualization layer, networking infrastructure, and hardware.	Customers are responsible for securing the operating system and applications running on their EC2 instances. This includes installing operating systems and software security patches, configuring host-based firewalls, managing security groups and network access control lists, implementing strong authentication and access controls, encrypting sensitive data where appropriate, monitoring system activity, and regularly updating applications and dependencies to protect against known vulnerabilities.	Failure to apply security patches, misconfigured security groups, weak access controls, or outdated applications can expose EC2 instances to unauthorized access, malware, ransomware, remote code execution attacks, and data breaches.
AWS CloudTrail	Provide and maintain the CloudTrail logging service and ensure service availability.	Customers are responsible for enabling CloudTrail logging, protecting log files from unauthorized access or deletion, monitoring account activity for suspicious events, reviewing logs regularly, and ensuring audit records are retained to support compliance, security monitoring, and incident response.	If CloudTrail is disabled, log files are deleted or modified, or audit logs are not monitored, organizations may fail to detect unauthorized activities, security incidents, or malicious changes on time. This can delay incident response.

Phase D: Independent Extension & Advanced Implementation

Phase Objective	Develop a basic threat model illustrating how misconfigurations could lead to compromise.
Estimated Time	~120 minutes
Evidence File(s)	PhaseD_Risk_Matrix.xlsx, PhaseD_threat_model_diagram.PNG

Actions Taken

I identified the main AWS assets used in this project, including Amazon S3, IAM, Amazon EC2, and AWS CloudTrail. For each asset, I identified a common security threat and evaluated its potential impact and likelihood.

I created a risk matrix with four columns: **Threat**, **Impact**, **Likelihood**, and **Mitigation**. The matrix included common cloud security risks such as publicly accessible S3 buckets, overly permissive IAM roles, insecure EC2 security groups, and disabled CloudTrail logging.

To visualize these risks, I created a threat model diagram in Draw.io. The diagram shows how an attacker could exploit cloud misconfigurations to gain unauthorized access to AWS resources. I also used AI to refine the mitigation descriptions and improve the clarity of the risk matrix before reviewing the final content.

AI Assistance Used

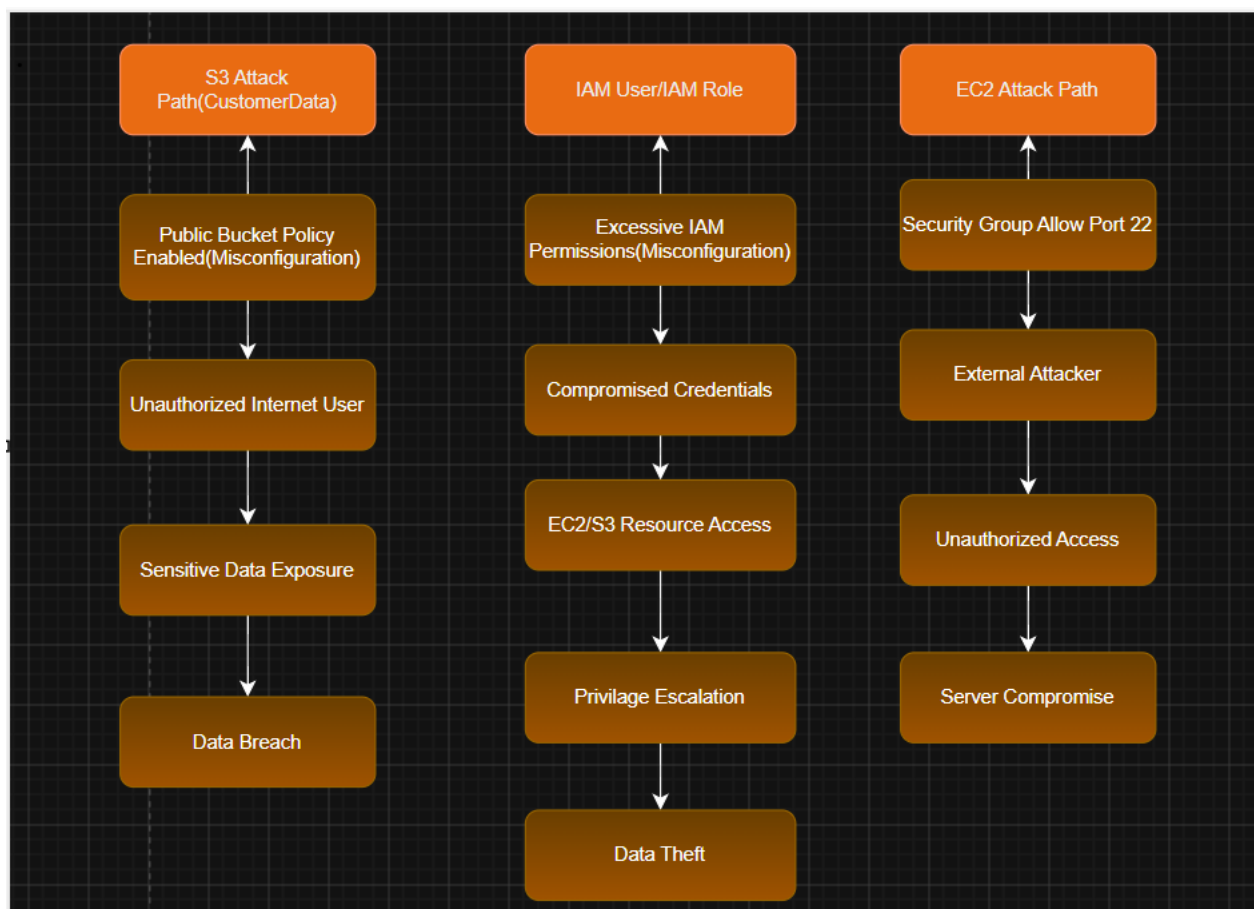
Prompt used

"Refine the mitigation descriptions in my AWS cloud security risk matrix and improve their clarity without changing the technical meaning."

Summary

ChatGPT helped improve the wording of the mitigation strategies to make them clearer and easier to understand. I reviewed each suggestion and confirmed that it aligned with AWS security best practices before including it in the report.

Figure 6. Threat model showing common attack paths caused by AWS misconfigurations.



Key Observations

This phase helped me understand how small configuration mistakes can lead to serious security risks in cloud environments. I observed that services such as IAM, S3, EC2, and CloudTrail are closely connected, and a weakness in one service can affect the security of others. Creating the risk matrix and threat model also showed how preventive controls, such as least privilege, secure bucket permissions, regular patching, and continuous logging, can reduce the likelihood of cloud attacks.

Evidence Produced

- PhaseD_Risk_Matrix.xlsx: Risk matrix showing identified threats, impact, likelihood, and mitigation strategies.
- PhaseD_threat_model_diagram.PNG: Threat model showing common attack paths caused by AWS misconfigurations.

Risk Matrix (Phase D Output)

[Reproduce or summarise your risk matrix here. If you produced it as a separate file (e.g. PhaseD_RiskMatrix.xlsx), include a screenshot of the matrix below and reference the full file as a deliverable. The risk matrix should cover at least 3 findings from this task.]

Cloud Asset / Resource	Threat / Risk	MITRE Ref.	Likelihood (1–5)	Impact (1–5)	Score	Rating
AWS S3 Bucket	Public Exposure via ACL misconfiguration	[T1530]	4	5	20	C
IAM Role	Privilege escalation via assume-role chain	[T1078]	4	5	20	C
AWS CloudTrail	Log tampering disabling visibility	[T1562.001]	3	4	12	H
Amazon EC2 Instance	Unpatched EC2 instance vulnerable to exploitation	[T1190]	3	4	12	H

Score = Likelihood × Impact | 1–4: Low | 5–9: Medium | 10–14: High | 15–25: Critical

Figure 7. Risk matrix summarizing identified cloud security threats, their likelihood, impact, and overall risk ratings.

A	B	C	D	E
Asset	Threat	Impact	Likelihood	Mitigation
Amazon S3 Bucket	Publicly exposed S3 bucket	Sensitive data may be disclosed, leading to data breaches, compliance violations, and reputational damage.(High)	High	Block public access, enable encryption, review bucket policies regularly, and apply the principle of least privilege.
IAM Users/Roles	Overprivileged IAM role	Attackers could gain unauthorized access, escalate privileges, and compromise AWS resources.(High)	High	Grant only required permissions, enable MFA, review IAM policies periodically, and remove unused accounts.
Amazon EC2 Instance	Misconfigured security group	Open ports may allow unauthorized remote access, malware infection, or exploitation of vulnerabilities.(High)	Medium	Restrict inbound rules, close unused ports, patch the operating system, and monitor network activity.
AWS CloudTrail	CloudTrail logging disabled	Security incidents may go undetected, delaying incident response and forensic investigations.(Medium)	Medium	Enable CloudTrail in all regions, protect log files, configure monitoring, and review audit logs regularly.

Phase E: Hosted Lab Integration

Phase Objective	Reinforce learning through guided labs simulating real-world scenarios.
Estimated Time	180 minutes
Evidence File(s)	PhaseE_labproof-PWNED-lab.PNG, PhaseE_labproof-TryHackMe.PNG

Actions Taken

I completed the TryHackMe Security Principles lab to review core cybersecurity concepts, including the CIA triad, authentication, authorization, least privilege, and defense in depth. The lab reinforced the security principles that are applied in AWS cloud environments.

I completed the PwnedLabs – Breach in the Cloud lab. I configured the AWS CLI using the temporary AWS credentials provided by the lab and verified access using the `aws sts get-caller-identity` command. I explored the AWS

environment using AWS CLI commands, examined IAM users, roles, S3 buckets, and CloudTrail logs, and investigated the simulated cloud security incident. During the investigation, I analyzed CloudTrail events, identified suspicious activities, and reviewed how IAM permissions and logging data could be used to understand the attack path. Screenshots of the completed labs were captured and saved as evidence.

Commands Used:

```
aws configure
```

```
aws sts get-caller-identity
```

```
aws iam list-users
```

```
aws iam list-roles
```

```
aws s3 ls
```

Key Observations

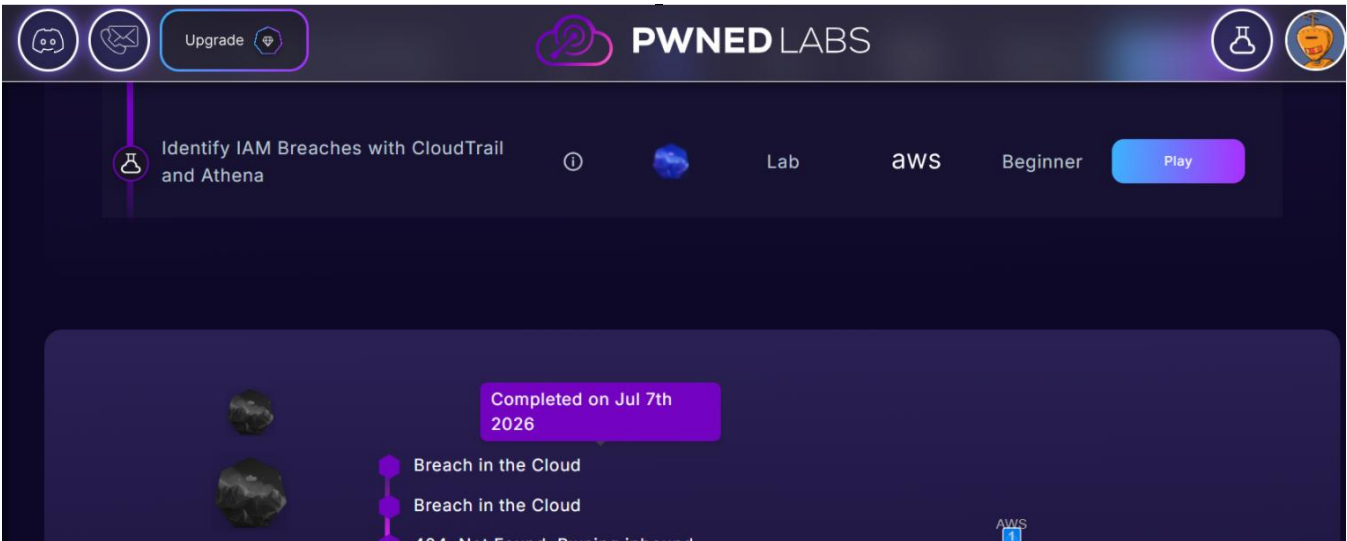
The hosted labs helped me apply cloud security concepts in realistic scenarios. I learned how CloudTrail logs can be used to investigate security incidents and how IAM permissions affect access to AWS resources. The exercises also showed the importance of monitoring, logging, and applying the principle of least privilege to reduce the impact of cloud attacks.

Evidence Produced

- [PhaseE_labproof-TryHackMe.PNG: Figure 8](#)



- [PhaseE_labproof-PWNED-lab.PNG: Figure 9](#)



Phase F: Documentation & Reporting

Phase Objective	Compile all findings into a professional report suitable for management review.
Estimated Time	120 minutes
Evidence File(s)	CSE_AWS_T001_Report.pdf) + ZIP archive of evidence files.

Actions Taken

I collected all evidence files from Phases A to E and organized them into a single project folder. I prepared the final report by documenting each phase in order, including the objectives, methodology, observations, and evidence. I added screenshots, the responsibility matrix, risk matrix, and threat model diagram to support my findings. Finally, I reviewed the report for accuracy, corrected formatting issues, and exported it as a PDF before creating a ZIP archive containing the report and all supporting evidence.

Key Observations

Preparing the final report helped me connect the work completed in each phase into one complete cloud security assessment. I observed that proper documentation is as important as the technical work because it clearly explains the environment, findings, risks, and recommendations. Organizing the evidence also made it easier to verify that every project requirement was completed.

Evidence Produced

- CSE_AWS_T001_Report.pdf — Final cloud security assessment report.
- CSE_AWS_T001_Report.zip — ZIP archive containing the report and all evidence files.
- PhaseA_Architecture_Diagram.png — Cloud architecture diagram.
- PhaseA_summary.pdf – Summay for shared Model .
- PhaseB_IAM_UsersList.png — Screenshot of IAM users.

- PhaseB_CloudTrail_Dashboard.png — Screenshot showing CloudTrail configuration.
- PhaseB_EC2_Instance.png — Screenshot of the running EC2 instance.
- PhaseB_S3_Bucket_CLI.png — Screenshot of aws command line for list of buckets.
- PhaseC_Responsibility_Matrix.xlsx — AWS Shared Responsibility Matrix.
- PhaseD_Risk_Matrix.xlsx — Risk assessment matrix.
- PhaseD_threat_model_diagram.png — Threat model created in Draw.io.
- PhaseE_labproof_TryHackMe.png — TryHackMe lab completion screenshot.
- PhaseE_labproof.png — PwnedLabs lab completion screenshot.

4.0 Findings

[Document every significant output of your task here. Findings for cloud security tasks include: misconfigurations, IAM policy weaknesses, overly permissive trust policies, public access exposures, encryption gaps, logging or monitoring blind spots, anomalous API activity, privilege escalation paths, and compliance deviations. If baseline checks were fully compliant with no anomalies, document that result as a finding and explain its significance.]

4.1 Findings Summary

#	Finding Title	Type / Category	Cloud Service / Resource	MITRE Ref.	Phase	Severity	Evidence Ref.
1	IAM User Initially Had Excessive Permissions	Access Control Weakness	IAM	T1078	Phase B	High	PhaseB_IAM_UsersList.png
2	CloudTrail Was Not Enabled by Default	Logging Gap	CloudTrail	T1562.001	Phase B	Medium	PhaseB_CloudTrail_Dashboard.png
3	Public S3 Bucket Could Expose Sensitive Data	Misconfiguration	Amazon S3	T1530	Phase D	High	PhaseD_risk_matrix.xlsx, PhaseD_threat_model_diagram.png
4	Overprivileged IAM Role Could Allow Unauthorized Access	IAM Policy Gap	IAM Roles	T1078	Phase D	High	PhaseD_risk_matrix.xlsx
5	EC2 Instance Requires Regular Security Hardening	Compliance Deviation	Amazon EC2	N/A	Phase C	Medium	PhaseB_EC2_Instance.png
6	CloudTrail Logs Should Be Protected Against Tampering	Logging Gap	CloudTrail	T1562.001	Phase D	Medium	PhaseD_threat_model_diagram.png
7	Hosted Labs Demonstrated Real Cloud Attack Techniques	Anomalous Activity	AWS Environment	T1078, T1530	Phase E	Informational	PhaseE_labProof_Pwned Labs.png

4.2 Detailed Finding Analysis

Finding 1: IAM User Initially Had Excessive Permissions

Finding Title	IAM User Initially Had Excessive Permissions
Type / Category	Access Control Weakness
Cloud Service / Resource	AWS IAM – Test User
Description	During the initial AWS setup, the test IAM user required additional permissions to perform AWS CLI commands such as listing S3 buckets. This highlighted the importance of assigning only the permissions required for a user's role. The IAM policy was reviewed to ensure access followed the principle of least privilege.
How Identified	AWS IAM Console, AWS CLI (aws iam list-users), and permission error messages when running CLI commands.
MITRE ATT&CK Reference	T1078 – Valid Accounts
Business / Security Impact	Excessive permission increases the risk of unauthorized access to AWS resources. If an attacker compromises the account, they could access or modify resources beyond what is necessary for the user's responsibilities
Severity / Impact	High
Evidence Reference	Figure 1. IAM test user created with the ReadOnlyAccess policy and S3ReadOnlyAccess

Finding 2: CloudTrail Logging Was Not Enabled by Default

Finding Title	CloudTrail Logging Was Not Enabled by Default.
Type / Category	Logging Gap
Cloud Service / Resource	AWS CloudTrail
Description	CloudTrail was not enabled at the beginning of the setup of environment. A multi-region trail was created so API activity could be recorded and monitored. This ensures security events are available for investigation and auditing.
How Identified	AWS CloudTrail Console and verification after enabling the trail.
MITRE ATT&CK Reference	T1562.001 – Impair Defenses: Disable or Modify Security Tools
Business / Security Impact	Without CloudTrail logs, suspicious API activity may go unnoticed. Missing logs make incident response and forensic investigations much more difficult
Severity / Impact	Medium
Evidence Reference	Figure 2. AWS CloudTrail configured for multi-region logging.

Finding 3: Public S3 Bucket Could Expose Sensitive Data

Finding Title	Public S3 Bucket Could Expose Sensitive Data.
Type / Category	Misconfiguration
Cloud Service / Resource	Amazon S3
Description	The threat model identified that an incorrectly configured S3 bucket with public access could expose sensitive files. Although no production data was stored in the lab, this represents one of the most common AWS security misconfigurations.
How Identified	Risk assessment, responsibility matrix, and threat model created during Phase D.
MITRE ATT&CK Reference	T1530 – Data from Cloud Storage Object.
Business / Security Impact	Public access to sensitive data can result in data breaches, regulatory violations, financial losses, and reputational damage.
Severity / Impact	High
Evidence Reference	Figure 6. Threat model showing common attack paths caused by AWS misconfigurations. Figure 7. Risk matrix summarizing identified cloud security threats, their likelihood, impact, and overall risk ratings.

Finding 4: Overprivileged IAM Role Could Allow Unauthorized Access

Finding Title	Overprivileged IAM Role Could Allow Unauthorized Access
Type / Category	IAM Policy Gap
Cloud Service / Resource	AWS IAM Role
Description	The threat model identified that assigning overly permissive IAM roles can allow users or applications to perform actions beyond their intended responsibilities. Permissions should follow the principle of least privilege and be reviewed regularly.
How Identified	Risk matrix and threat model developed in Phase D. IAM permission review in the AWS Console.
MITRE ATT&CK Reference	T1078 – Valid Accounts
Business / Security Impact	Overprivileged roles increase the risk of privilege escalation, unauthorized resource modification, and data exposure if an account is compromised.
Severity / Impact	High
Evidence Reference	Figure 6. Threat model showing common attack paths caused by AWS misconfigurations.

Figure 7. Risk matrix summarizing identified cloud security threats, their likelihood, impact, and overall risk ratings.

Finding 5: EC2 Instance Requires Security Hardening

Finding Title	EC2 Instance Requires Security Hardening
Type / Category	Compliance Deviation
Cloud Service / Resource	Amazon EC2
Description	A Free Tier EC2 instance was deployed successfully for testing. The instance should be regularly patched, monitored, protected with restrictive security groups, and configured according to AWS security best practices.
How Identified	AWS EC2 Console during environment setup and security review.
MITRE ATT&CK Reference	N/A
Business / Security Impact	Unpatched or poorly configured EC2 instances can become entry points for attackers, resulting in unauthorized access or malware infections.
Severity / Impact	Medium
Evidence Reference	Figure 3 – PhaseB_EC2_Instance.png

Finding 6: CloudTrail Log Files Should Be Protected Against Tampering

Finding Title	CloudTrail Log Files Should Be Protected Against Tampering
Type / Category	Logging Gap
Cloud Service / Resource	Amazon CloudTrail
Description	CloudTrail provides detailed records of AWS API activity. Log files should be encrypted, protected from unauthorized modification, and stored securely to preserve their integrity during investigations.
How Identified	CloudTrail configuration review and Phase D threat modelling exercise.
MITRE ATT&CK Reference	T1562.001 – Impair Defenses
Business / Security Impact	If attackers can modify or delete log files, security teams may lose important evidence needed to investigate incidents and determine the attack timeline
Severity / Impact	Medium
Evidence Reference	Figure 6. Threat model showing common attack paths caused by AWS misconfigurations. Figure 2. AWS CloudTrail configured for multi-region logging.

Finding 7:

Finding Title	Hosted Labs Demonstrated Common Cloud Attack Techniques
Type / Category	Informational / Security Awareness
Cloud Service / Resource	PwnedLabs & TryHackMe
Description	The hosted labs simulated realistic cloud security scenarios involving IAM permissions, CloudTrail analysis, and cloud resource investigation. These exercises reinforced practical incident investigation skills and demonstrated how cloud attacks can be detected and analyzed.
How Identified	Completion of TryHackMe Security Principles and PwnedLabs Breach in the Cloud labs.
MITRE ATT&CK Reference	T1078 – Valid Accounts, T1530 – Data from Cloud Storage Object
Business / Security Impact	The exercises improved understanding of cloud attack techniques and strengthened the ability to identify, investigate, and respond to security incidents in AWS environments.
Severity / Impact	Informational
Evidence Reference	Figure 8 – PhaseE_labProof_TryHackMe.png, Figure 9 – PhaseE_labProof_PwnedLab.png

5.0 Risk Assessment & Cloud Threat Model

[This section is required for all tasks. Complete both sub-sections using the outputs from your Phase D threat modelling work and your findings from Section 4.]

5.1 Cloud Security Control Gaps & Defensive Coverage

Finding / Gap	Observable / Trigger in Cloud Telemetry	Recommended Control	AWS Service / Tool for Implementation
IAM User Initially Had Excessive Permissions	Unauthorized or unexpected AttachUserPolicy, PutUserPolicy, or CreateAccessKey events recorded in CloudTrail.	Apply least-privilege IAM policies, review permissions regularly, and enable IAM Access Analyzer to identify excessive permissions.	AWS IAM, IAM Access Analyzer, AWS CloudTrail
CloudTrail Logging Was Not Enabled by Default	Missing CloudTrail events or absence of API activity logs for AWS resources.	Enable a multi-region CloudTrail trail, encrypt log files, and enable log file validation to protect audit logs.	AWS CloudTrail, Amazon S3, AWS KMS

Finding / Gap	Observable / Trigger in Cloud Telemetry	Recommended Control	AWS Service / Tool for Implementation
Public S3 Bucket Could Expose Sensitive Data	Public GetObject requests, changes to bucket policies, or ACL modifications recorded in CloudTrail.	Public GetObject requests, changes to bucket policies, or ACL modifications recorded in CloudTrail.	Amazon S3, AWS Config, IAM Access Analyzer
Overprivileged IAM Role Could Allow Unauthorized Access	Unusual AssumeRole events or privileged API calls from unexpected users or services.	Apply least-privilege role policies, review trust relationships, and monitor AssumeRole activity with CloudTrail and CloudWatch.	AWS IAM, AWS CloudTrail, Amazon CloudWatch
EC2 Instance Requires Security Hardening	Unexpected inbound connections, security group changes, or unusual instance activity recorded in CloudTrail and VPC Flow Logs.	regularly patch EC2 instances, restrict security group rules, enable monitoring, and use Systems Manager for patch management.	Amazon EC2, AWS Systems Manager, VPC Flow Logs, Amazon CloudWatch

5.2 Risk Exposure Over Time

Risk 1 — [Title]: Public S3 Bucket Exposure

If an S3 bucket is accidentally configured for public access, sensitive files could be accessed by unauthorized users. In this project, public bucket access was identified as a high-risk misconfiguration during the threat modelling exercise. If left unaddressed, it could lead to data breaches, compliance violations, and reputational damage.

Risk 2 — [Title]: Overprivileged IAM Permissions

Granting users or roles more permissions than necessary increases the risk of unauthorized access. During this project, IAM permissions were reviewed to follow the principle of least privilege. If excessive permissions remain, a compromised account could modify AWS resources, access sensitive data, or perform privilege escalation.

Risk 3 — CloudTrail Logging Disabled or Misconfigured

CloudTrail provides the audit logs needed to detect and investigate security incidents. During the environment setup, CloudTrail was enabled to record API activity across the AWS account. If logging is disabled or misconfigured, security teams may not detect malicious actions or have sufficient evidence to investigate an incident

6.0 Recommendations

6.1 Technical Remediation

#	Finding	Specific Remediation Action (with Policy / CLI where applicable)	Priority	Standard / Reference
1	IAM User Initially	Review all IAM users and remove unnecessary permissions. Attach only least-privilege policies based on job responsibilities. Enable IAM Access Analyzer to detect excessive permissions and review permissions regularly.	Immediate	CIS AWS Foundations v1.5 Control 1.16, NIST SP 800-53 AC-6, ISO 27001 A.5.15
2	CloudTrail Logging Was Not Enabled by Default	Enable a multi-region CloudTrail trail. Store logs in a dedicated S3 bucket, enable log file validation, and encrypt logs using AWS KMS. Configure CloudWatch alarms to notify administrators if CloudTrail is disabled.	Immediate	CIS AWS Foundations v1.5 Control 3.x, NIST SP 800-53 AU-2, ISO 27001 A.8.15
3	Public S3 Bucket C	Enable S3 Block Public Access at both the account and bucket levels. Remove any public ACLs or bucket policies and allow access only to authorized IAM identities. Example CLI: <code>aws s3api put-public-access-block --bucket <bucket-name> --public-access-block-configuration BlockPublicAcls=true,IgnorePublicAcls=true,BlockPublicPolicy=true,RestrictPublicBuckets=true</code>	Immediate	CIS AWS Foundations v1.5 Control 3.x, NIST SP 800-53 AU-2, ISO 27001 A.8.15
4	Overprivileged IAM Role Could Allow Unauthorized Access	Review IAM role trust policies and remove unnecessary permissions. Use least-privilege policies, enable IAM Access Analyzer, and require MFA for privileged access. Review roles periodically to ensure permissions remain appropriate.	Short-term	CIS AWS Foundations v1.5 Control 3.x, NIST SP 800-53 AU-2, ISO 27001 A.8.15
5	EC2 Instance Requires Security Hardening	Apply operating system updates regularly, restrict inbound security group rules, disable unnecessary services, and use AWS Systems Manager Patch Manager for automated patching. Enable CloudWatch monitoring for instance health and security events.	Short-term	CIS AWS Foundations v1.5 Control 3.x, NIST SP 800-53 AU-2, ISO 27001 A.8.15

6.2 Strategic & Operational Improvements

1. Strengthen IAM Governance

Implement a least-privilege access model for all IAM users and roles. Require periodic permission reviews, enable MFA for privileged accounts, and use IAM Access Analyzer to identify overly permissive policies before they can be exploited.

2. Improve S3 Security Controls

Enforce S3 Block Public Access across the AWS account and require bucket encryption by default. Configure AWS Config rules to continuously monitor bucket permissions and generate alerts whenever public access is detected.

3. Enhance Logging and Monitoring

Enable CloudTrail across all regions and store logs in a dedicated, encrypted S3 bucket with log file validation enabled. Configure Amazon CloudWatch alarms to notify administrators of suspicious activities such as unauthorized API calls, policy changes, or attempts to disable logging.

4. Establish Continuous Security Monitoring

Enable AWS Security Hub and Amazon GuardDuty to continuously monitor the AWS environment for misconfigurations and potential threats. Regularly review findings and remediate high-risk issues as part of routine security operations.

5. Perform Regular Security Reviews

Schedule periodic security assessments, including IAM permission audits, S3 bucket access reviews, and CloudTrail configuration verification. Document findings and update security controls to maintain compliance with AWS security best practices and organizational policies.

7.0 Reflection

Q1 How does misunderstanding the shared responsibility model increase business risk?

If a company does not understand the shared responsibility model, important security tasks can be missed. In this lab, I learned that AWS protects the cloud, but customers must secure their own services and data. A mistake like a public S3 bucket or weak IAM permissions can lead to data exposure.

Q2 Which part of your own configuration posed the highest potential impact if mismanaged?

The IAM configuration had the biggest impact if it was set up incorrectly. Giving users or roles too many permissions could allow unauthorized access to AWS resources. This lab showed me why least privilege is important and why permissions should be reviewed regularly.

Q3 How would you communicate shared responsibility concepts to non-technical executives?

I would explain that AWS and the customer both have security responsibilities. AWS protects the cloud infrastructure, while the customer protects their data, users, and settings. When both sides do their part, the cloud environment is much more secure. Also providing example of Office rented.

Q4 What proactive monitoring controls could reduce human error in cloud configurations?

Tools like CloudTrail, AWS Config, Security Hub, and IAM Access Analyzer can help find security issues early. They can alert administrators when there are risky changes or misconfigurations. Regular reviews and monitoring help prevent small mistakes from becoming serious security problems.

8.0 Deliverables Submission Checklist

✓	Phase	Deliverable Description	Your Filename	Status
☐	Phase A	Upload your diagram (shared_responsibility_diagram.png) and summary (phaseA_summary.pdf).	<i>PhaseA_SharingResponsibilityDiagram, PhaseA_Summary</i>	<i>Completed</i>
☐	Phase B	Screenshots of IAM user list, CloudTrail dashboard, EC2 instance page (phaseB_evidence.zip).	PhaseB_CloudTrail_Dashboard.PNG, PhaseB_EC2_Instance.PNG, PhaseB_IAM_UserList.PNG, PhaseB_S3_Buckets_CLI.PNG	<i>Completed</i>
☐	Phase C	Upload spreadsheet (responsibility_matrix.xlsx).	PhaseC_Responsibility_Matrix.xlsx,	<i>Completed</i>
☐	Phase D	Upload risk_matrix.xlsx and threat_model_diagram.png.	PhaseD_Risk_Matrix.xlsx, PhaseD_threat_model_diagram.PNG	<i>Completed</i>
☐	Phase E	Screenshots of completion dashboards (phaseE_labproof.png).	<i>PhaseE_labproof-PWNED-lab.PNG, PhaseE_labproof-TryHackMe.PNG</i>	<i>Completed</i>
☐	Report	This report — exported as PDF using the naming convention below	CSE_AWS-T-001-02_Report.pdf	<i>Completed</i>

9.0 Additional Observations

- No additional observations for this task.

Declaration & Sign-Off

I confirm that:

- All work in this report is my own except where AI usage is explicitly declared in Section 2.3.
- All technical activities were performed exclusively within the authorized cloud environments listed in Section 1.3.
- No external AWS accounts, production environments, third-party cloud resources, or real customer data were accessed during this task.
- All AWS credentials, access keys, session tokens, and account identifiers used during this task have been redacted in evidence files before submission.
- I have reviewed this report for accuracy, completeness, and professional presentation before submission.

Intern Full Name	Signature	Date of Submission
[Redacted]	[Redacted]	07-08-2026

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